

Clinically anchored multimodal survival prediction under missing, shifted and shortcut-prone evidence in head and neck cancer

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Abstract

Multimodal prognostic models are usually optimised to combine every recorded input, yet clinical data can be absent, technically unusable, shifted across platforms or predictive for the wrong reason. We developed PATTERN-Surv-HN, a clinically anchored framework for postoperative head and neck squamous cell carcinoma survival prediction that learns only residual evidence from an unordered set of usable modalities. A calibration bridge separates transport of risk ranking from transport of absolute risk, and a cross-fitted router selects fusion, clinical fallback, rank-only output or abstention. The evidence programme uses one primary development ecosystem and three prespecified falsification challenges: natural and unseen modality combinations in HANCOCK, RNA-sequencing-to-microarray probability transport, and present-but-invalid radiomic controls in RADCURE, followed by outcome-untouched confirmation. In 610 eligible HANCOCK development patients with 173 deaths, the completed clinical-pathological anchor achieved a mean 24-month IPCW Brier score of 0.1247, Uno's C of 0.6442, time-dependent AUC of 0.6620, calibration-in-the-large of **-0.0037** and calibration slope of 0.9367 under repeated nested cross-fitting. **[TBD: Insert the primary full-coverage effect, worst-pattern regret and matched-coverage routing result after model freeze.]** These analyses test a clinically consequential alternative to compulsory fusion: optional modalities should alter prognosis only when their patient-specific incremental evidence and probability scale are credible.

Keywords: head and neck squamous cell carcinoma, multimodal learning, survival analysis, missing modalities, calibration, selective prediction, trustworthy artificial intelligence

Living-manuscript status (14 August 2026). Only the HANCOCK data contract and clinical-pathological V0 anchor have completed the current workflow. Red **[TBD:]** fields require frozen analysis or author metadata. Previously outcome-seen cohorts are post-lock exploratory and are not newly untouched validation.

1 Introduction

Head and neck squamous cell carcinoma (HNSCC) comprises clinically and biologically heterogeneous diseases whose outcomes reflect anatomical site, tumour and nodal burden, human papillomavirus-related biology, smoking exposure, pathological features, treatment and host condition. Molecular studies have further resolved genomic and transcriptomic phenotypes not captured by stage alone [1–4]. Prognostic models therefore increasingly combine clinical, pathological, imaging, blood, tissue-microenvironment and molecular data; resources such as HANCOCK and RADCURE, together with recent HNSCC fusion studies, make this technically feasible [5–7]. The clinically relevant question, however, is not simply whether more modalities improve an average performance metric. It is whether optional evidence can safely alter a prediction that must still be available when those modalities are absent or unreliable.

A single distinction organises this problem: recorded data are not necessarily trustworthy evidence. First, a modality may be missing because it was not acquired, was structurally unavailable or failed preprocessing; naturally occurring combinations need not resemble random masking. Second, a modality may transport enough signal to preserve patient ranking while the baseline risk and probability scale shift across centres or assay platforms. Third, a modality may be present and apparently predictive but encode tumour volume, site, device, batch or processing shortcuts rather than the intended biology. Complete-case fusion and missing-value imputation conflate these states, although they require different clinical responses.

Classical Cox models provide interpretable, censoring-aware estimation but generally assume a fixed covariate vector [8, 9]. Random survival forests and neural survival models accommodate non-linearity, yet do not by themselves decide whether an optional modality should be used [10–12]. Recent incomplete-modality methods reconstruct or distil missing representations, use modality dropout or prompts, align heterogeneous embeddings, or aggregate available inputs with attention [13–15]. These studies establish that survival learning can remain feasible as modalities disappear. The unresolved translational gap is broader: natural acquisition and usability, combinations absent during training, exact clinical fallback, present-but-invalid controls, transport of calibrated probabilities and selective reporting must be evaluated within one governed prediction pathway.

Discrimination alone is insufficient for that pathway. Concordance and time-dependent AUC describe ranking, whereas a statement such as “24-month mortality

risk is 35%” additionally requires a transportable baseline hazard and calibration relationship. A model can preserve ordering under domain shift while becoming systematically overconfident. Likewise, a selective model can appear safer simply by excluding difficult patients, and average performance can hide severe failure in a rare modality pattern. Evaluation must therefore retain fallback patients in the full-coverage estimand, compare selective policies on identical retained patients, report calibration and coverage by pattern, and use negative controls to test whether apparent value is specific to the intended modality [16–21].

We developed PATTERN-Surv-HN around one framework, three falsification challenges and four supported outputs. A postoperative clinical-pathological model supplies a prediction for every eligible patient. Available non-clinical modalities form an unordered set and can contribute only residual evidence beyond this anchor. HANCOCK tests natural absence, acquired-but-unusable inputs and unseen combinations; TCGA–GEO transfer tests whether ranking can survive when absolute risk does not; and RADCURE tests whether a present modality remains more informative than volume, permutation and random-feature controls. A calibration bridge and strictly cross-fitted value–reliability router then select FUSE, FALLBACK, RANK-ONLY or ABSTAIN. The primary claim is deliberately narrow: conditional evidence use should improve full-population probabilistic accuracy over a strong late-fusion comparator without increasing worst-pattern harm relative to the clinical anchor. Strong external-generalisation claims additionally require a frozen, outcome-untouched postoperative cohort.

2 Results

2.1 One framework is evaluated through three falsification challenges

The intended-use population comprises adults with HNSCC whose prognosis is estimated immediately after definitive surgery and standard pathology review, before adjuvant treatment is used as a predictor. Overall survival is measured from the first treatment date to the last information date; death from any cause is the event, and 730.5 days is the primary horizon. Rather than pool heterogeneous cohorts under a single external-validation label, we assigned each dataset a falsification role. HANCOCK is the primary development ecosystem for natural acquisition, usability and set learning; TCGA–GEO transfer isolates probability-scale failure; RADCURE probes shortcut-prone observed evidence; and only a future outcome-untouched postoperative cohort can support final confirmation (Table 1).

The result narrative follows a gated evidence ladder. Level 1 establishes the data contract and full-coverage clinical anchor. Level 2 tests incremental benefit against strong simple and recent incomplete-modality comparators. Level 3 attempts to falsify reliability through natural, unseen, shifted and corrupted evidence. Level 4 requires frozen outcome-untouched confirmation. A favourable secondary mechanism cannot rescue failure of the primary full-coverage or worst-pattern estimand. At the manuscript date, only Level 1 is complete; later levels remain prespecified, auditable result slots rather than inferred findings.

Table 1 Cohorts, falsification roles and claim boundaries.

Cohort	Prediction setting	Size/events	Modalities	Primary role	Governance/claim boundary
HANCOCK training	Postoperative	610/173	Clinical-pathological, blood, ICD, TMA	Development, nested cross-fitting, natural-pattern and corruption studies	PATTERN development; official-test outcomes not used for V0
HANCOCK official test	Postoperative	152/40	Same ecosystem	Fixed held-out exploratory evaluation	Outcomes seen in Phase 6; post-lock exploratory
TCGA-HNSC	Cohort-specific OS	[TBD; count]	Clinical, RNA-seq	Transcriptomic source development	Source model only; timing difference explicit
GSE6988	Retrospective external	244/78	Clinical, microarray	Ranking; calibration transfer	Outcome-seen; post-lock exploratory
GSE4113	On-ticity sensitivity	97/51	Clinical, microarray	Sensitivity analysis	Outcome-seen; restricted case mix
RADCURE	Pretreatment radiotherapy	626/110	Clinical, volume, CT radionics	Shortcut/invalid-modality replication	Outcome-seen; mechanism study
Future external/temporal	Frozen postoperative use	[TBD; cohort]	Anchor plus optional modality	Final confirmation	Must remain outcome-untouched until complete freeze

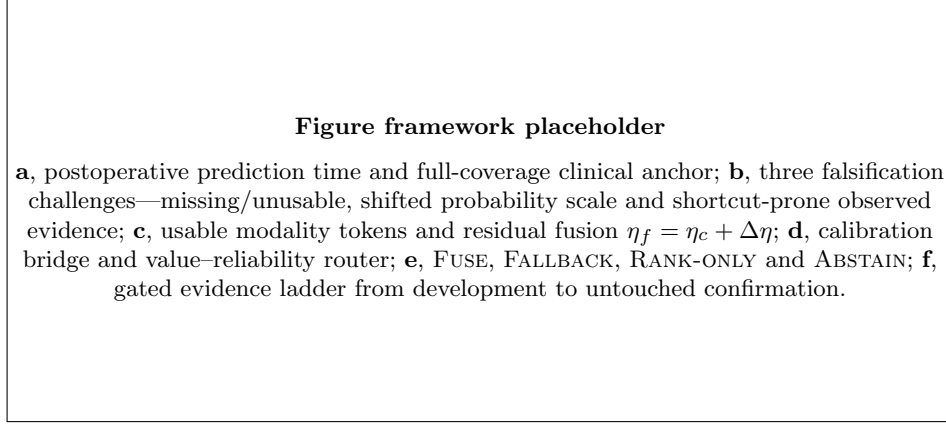


Fig. 1 One framework, three falsification challenges and four supported outputs. Final artwork should show clinical fallback as an exact computational path and distinguish evidence roles from cohort names.

2.2 Natural acquisition and usability create sparse modality combinations

The full HANCOCK resource contains 763 patients. The official training split contains 611 patients, of whom one was excluded for non-positive postoperative duration, leaving 610 patients and 173 deaths for development. A deterministic contract allocated 489 observations to training and 122 to calibration, while the official test set of 152 remained outside V0 development.

For blood, 693 patients had acquired data, 683 had usable data, 70 were absent and 10 were acquired but unusable. Among usable blood records, 604 were complete and 79 internally partial. Across blood, ICD and TMA, raw acquired-pattern counts for binary codes 001, 010, 011, 100, 101, 110 and 111 were 7, 4, 59, 1, 43, 22 and 627. Usable-pattern counts were 8, 5, 67, 1, 42, 21 and 619. Thus, acquisition is not proof of model-ready information, and separate models are infeasible for several rare combinations.

2.3 The clinical-pathological anchor establishes a full-coverage reference

The anchor used age, sex, smoking, primary site, grade, p16, resection, pathological T category and pathological N category. Adjuvant treatment and post-prediction variables were prohibited. An elastic-net Cox model used five seeds, five outer folds and

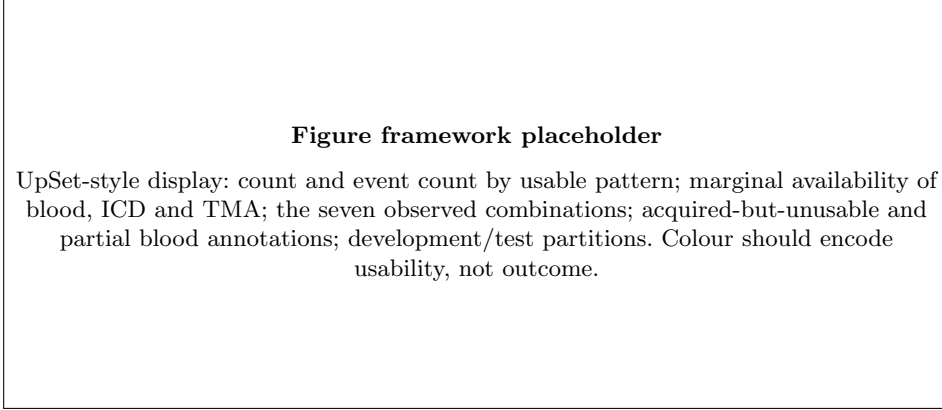


Fig. 2 Natural acquisition and usability patterns in HANCOCK. Final artwork will add development-only event counts and outcome-blind quality indicators.

Table 2 Completed repeated nested cross-fitting of the HANCOCK V0 anchor.

Metric	Mean	SD across seeds
IPCW Brier at 24 months	0.1247	0.0015
Harrell C	0.6230	0.0144
Uno C at 24 months	0.6442	0.0138
Time-dependent AUC	0.6620	0.0144
Calibration-in-the-large	-0.0037	0.0149
Calibration slope	0.9367	0.0792

three inner folds. All imputation, coding, scaling, penalty selection and Breslow baseline estimation were confined to the training fold. Repeated out-of-fold predictions covered 3,050 patient-seed instances without official-test outcomes.

Mean 24-month IPCW Brier score was 0.1247 ± 0.0015 . Harrell concordance was 0.6230 ± 0.0144 , Uno 24-month concordance was 0.6442 ± 0.0138 , and 24-month AUC was 0.6620 ± 0.0144 . Calibration-in-the-large was -0.0037 ± 0.0149 and calibration slope was 0.9367 ± 0.0792 . Two formal reruns produced identical out-of-fold hashes. These findings indicate moderate internal discrimination and approximately acceptable average calibration, sufficient to operationalise V0 as a post-lock exploratory fallback reference. They do not establish multimodal benefit, missing-modality robustness, external generalisation or clinical utility.

2.4 Residual fusion is judged by full-coverage benefit and worst-pattern harm

The primary estimand is the 24-month IPCW Brier difference between the complete PATTERN policy and prespecified conventional late-fusion comparator C3. Patients routed to clinical fallback remain in the numerator and denominator using their anchor

Table 3 Primary HANCOCK full-coverage comparison. Red fields require frozen analysis.

Model/policy	Brier ₂₄	IBS	Uno C ₂₄	AUC ₂₄	Cal. int.	Cal. slope	Interpretation
KM constant risk	[TBD: value]	[TBD:]	—	—	[TBD:]	—	Outcome-frequency reference
V0 clinical anchor	0.1247±0.0015	[TBD:]	0.6442±0.0138	0.6620±0.0144	−0.0037	0.9367	Completed internal OOF
C3 late fusion	[TBD: CI]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	Primary simple comparator
HAF reproduction	[TBD: CI]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	Recent HANCOCK benchmark
Deep Sets residual	[TBD: CI]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	Minimum set model
Set Transformer residual	[TBD: CI]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	Complexity only if justified
PATTERN policy	[TBD: CI]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]	Routing and fallback included

Table 4 Pattern and stress-scenario reporting matrix.

Scenario	Population	<i>n</i> /events	ΔBrier ₂₄ vs anchor	Cal. slope	Fallback/abstain	Interpretation
Natural complete	HANCOCK	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Natural incomplete	Supported cells/pooled	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Unseen combination	Leave-one-pattern-out	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Block dropout	Complete cases minus one input	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
MAR-like missingness	Covariate-dependent masking	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Corrupted present modality	Noise/shift/permutation	[TBD:]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Worst supported pattern	Prespecified maximum regret	[TBD:]	[TBD:]	[TBD:]	[TBD:]	Co-primary safety estimand

probability. Rank-only or abstained observations follow the frozen policy and cannot disappear from a full-coverage claim. The co-primary safety estimand is the largest pattern-specific excess Brier score relative to the anchor.

A publishable positive interpretation requires a clinically relevant full-coverage improvement with confidence intervals excluding unacceptable harm and non-inferior worst-pattern regret. If Deep Sets or Set Transformer does not outperform C3, that negative finding will be reported and architectural superiority will not be claimed.

2.5 Natural absence, unseen combinations and corrupted inputs probe distinct failures

Natural-pattern analysis quantifies performance in combinations generated by routine acquisition and usability. Seen/unseen experiments hold out one supported combination during training and test compositional generalisation. Corruption experiments retain availability while degrading content through noise, location/scale shift, within-feature permutation, across-patient permutation or modality replacement. A model can handle an absent token yet still trust a corrupted present token; the experiments are therefore not interchangeable.

The principal robustness display will report full-to-subset degradation, seen–unseen gaps, calibration and safety regret. Pattern-specific estimates will ordinarily require at least 30 patients and 10 events; rarer cells will be pooled hierarchically or described without definitive claims. The decisive criterion is whether the policy avoids materially exceeding the anchor in its worst supported pattern.

2.6 Cross-platform transport separates stable ranking from valid probability

TCGA-HNSC to GSE65858 and GSE41613 transfer is a mechanism study of cross-platform failure rather than untouched validation. A source ranking representation will be frozen, then probability recovery examined with zero target events and calibration samples containing 5, 10, 20 and 40 events. Strategies include the source Breslow

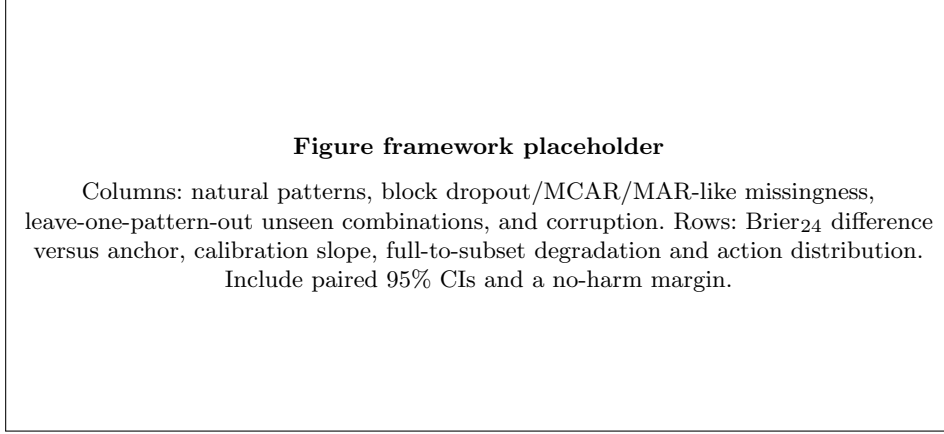


Fig. 3 Robustness to incomplete and unreliable modality sets. All estimates are [TBD: frozen analysis].

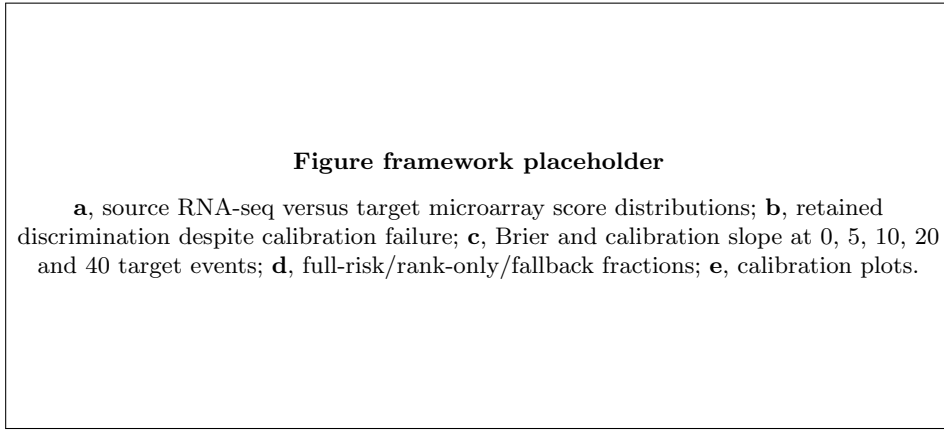


Fig. 4 Separating transport of ranking from transport of absolute risk. Results are [TBD: post-lock exploratory frozen analysis].

baseline, target baseline hazard, intercept-only, intercept-plus-slope, isotonic calibration and the hierarchical bridge. Both GEO outcomes were previously examined, so all results remain post-lock exploratory.

A useful model may preserve concordance while source probabilities are unusable. The correct output is then RANK-ONLY, not a precise unsupported probability. Figure 4 will plot discrimination, Brier score, calibration slope and the fraction eligible for full-risk reporting as target events accumulate. Improvement at small event counts will be interpreted with uncertainty and effective sample size [22, 23].

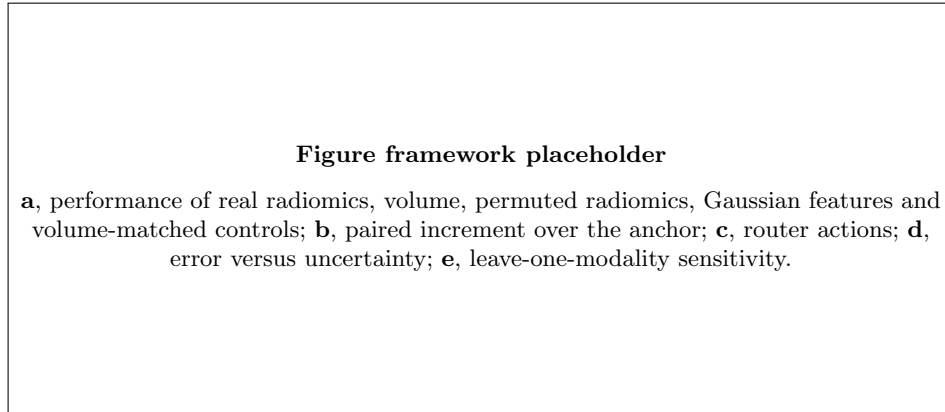


Fig. 5 Present-but-invalid modality controls and router behaviour. Values are [TBD: post-lock exploratory replication].

2.7 Negative controls test whether observed modalities contribute specific evidence

RADCURE provides a complementary failure mode: a radiomic vector exists but apparent performance may derive from tumour volume, centre, preprocessing or high-dimensional structure. The replication compares clinical only, volume only, clinical plus volume, radiomics only, clinical plus radiomics and full fusion with patient-permuted radiomics, Gaussian random features, volume-matched random controls, and scaled/corrupted radiomics. The question is whether the learned increment is more specific than appropriate negative controls.

Earlier frozen project analyses found that original RADCURE radiomics did not clearly outperform shuffled or randomised controls, while gating did not consistently improve forced fusion. These observations motivated this protocol but do not show that the untrained PATTERN router succeeds. The new analysis is supportive only if the router reduces use of invalid controls without suppressing reproducible real-modality value, on matched patients rather than through denominator changes.

2.8 Reliability routing is evaluated by regret at matched coverage

Risk-coverage curves will be reported at 100%, 95%, 90% and 80% coverage. Every lower-coverage comparison will evaluate the candidate and comparator on the identical retained patients. Coverage will be stratified by age, sex, site, p16 and modality pattern so that lower error cannot be achieved by disproportionately excluding a clinically important group. Actions will be accompanied by event counts and calibration uncertainty.

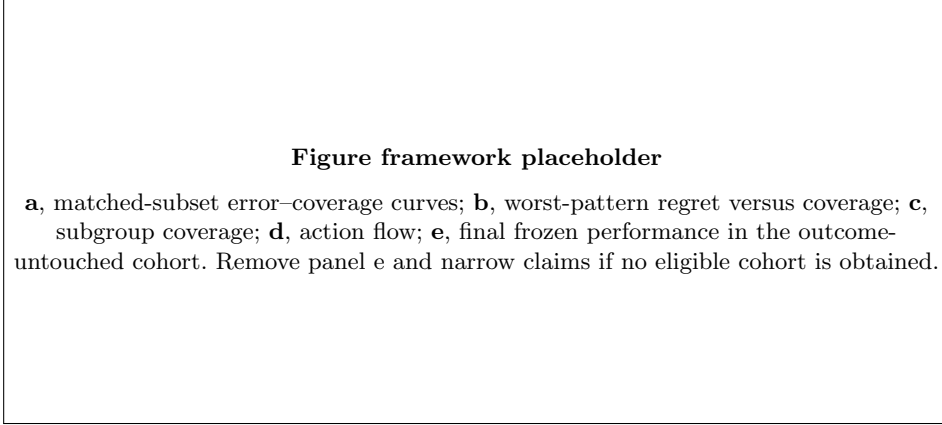


Fig. 6 Selective prediction, safety regret and confirmatory validation. Values are [TBD: frozen analysis].

3 Discussion

This study is designed around a simple clinical proposition: additional data should modify a stable prediction only when their incremental evidence is credible. That differs from maximising an average C-index under a complete-modality matrix. In HNSCC, the operational problem includes naturally missing tests, internally incomplete measurements, combinations absent from training, platform-dependent probability scales and apparently informative high-dimensional controls. PATTERN-Surv-HNtreats these as related forms of uncertain evidence.

Recent work establishes both the opportunity and the benchmark. HNSCC fusion models combine clinical, radiomic and deep imaging information for prognosis and treatment-response assessment, while prompt distillation, heterogeneous aligned fusion and attention-based aggregation address incomplete multimodal survival data [7, 13–15]. The proposed advance is therefore not merely another fusion block. The distinguishing claim is that a system can recognise when optional evidence should not modify an otherwise available clinical prediction, and that this behaviour can be falsified under natural selection of modalities, probability shift and negative controls.

The completed V0 analysis establishes the first component. The postoperative anchor achieved moderate discrimination and near-unit average calibration slope in repeated nested cross-fitting. Its role is not to be declared clinically sufficient or externally valid. It provides a reference for every eligible patient, defines residual learning and makes fallback testable. Without a stable anchor, a router chooses only between differently unstable models.

The residual construction constrains the claim. The fused score is $\eta_c + \Delta\eta$, so optional modalities explain information beyond the anchor rather than relearn the outcome. Exact fallback when no usable token is present prevents numerical drift from masquerading as clinical-only prediction. Set encoding avoids assigning meaning to modality order and permits combinations not enumerated as separate models. Yet

permutation invariance is not robustness: a set model may still overfit rare patterns or trust corrupted tokens, hence the unseen-combination and present-but-invalid tests.

Separating ranking from probability is central. Baseline hazard, event incidence, follow-up and assay calibration can shift independently of rank ordering. The RANK-ONLY action makes this visible. An explicit relative-risk statement is more honest than a precisely formatted probability unsupported by target calibration. Conversely, a flexible calibrator fitted to few events may improve apparent fit but remain unstable, so event-budget curves must include uncertainty.

Negative controls raise the evidential standard. If real radiomics perform no better than patient-permuted or random features after accounting for clinical variables and volume, specificity has not been demonstrated. A reliable router should react, but it must not be trained on final external outcomes. Cross-fitted value labels, corruption probes and a separately frozen confirmation cohort are therefore integral, not optional decoration.

Several outcomes would change the final interpretation. If C3 matches the set models, we will prefer the simpler comparator. If fusion improves average Brier score but harms a supported pattern, the safety hypothesis fails. If routing lowers error only by reducing coverage, matched-subset and subgroup analyses will expose the trade-off. If ranking transfers but calibration cannot be restored, conclusions will be limited to stratification. Without an untouched cohort, the study can support a post-lock exploratory framework but not strong external-generalisation claims.

Strengths include explicit prediction time and prohibited post-treatment variables; cross-fitted preprocessing, model selection and router supervision; distinction between missing, unusable and corrupted inputs; full-population primary estimands; negative controls; and emphasis on paired effect sizes with uncertainty. Reporting follows TRIPOD+AI, with PROBAST+AI structuring bias and applicability assessment [24, 25].

Limitations are substantial. All current cohorts are retrospective and differ in eligibility, treatment era, outcome construction and platform. Natural patterns may encode selection processes that cannot be fully adjusted. Several patterns are too rare for independent evaluation. V0 is internal and post-lock exploratory. RADCURE and both GEO cohorts are outcome-seen and cannot serve as new confirmation. Calibration requires enough target events, and action thresholds lack prospective workflow validation. Decision-curve analysis describes potential net benefit but cannot establish clinical utility without a defined decision and impact study [26]. No clinician interaction, patient acceptability, workload, fairness-impact or prospective silent-deployment study has occurred.

Translation should proceed in stages: freeze a parsimonious model and thresholds; confirm technical and calibration performance in an outcome-untouched temporal or external cohort; perform prospective silent validation with real-time missingness and quality logging; predefine responses to fallback, rank-only and abstention; then evaluate decision impact. Until then, PATTERN-Surv-HN is a reliability-oriented research framework, not a deployable medical device.

4 Methods

4.1 Study design and reporting framework

We designed a retrospective, multi-cohort prognostic modelling study with separately governed development and a required future confirmatory stage. The protocol label is `post_lock_exploratory`, because outcomes in several project cohorts had been accessed before the unified method was specified. Frozen Phase 3–6 configurations, implementations and aggregate results are not modified. Patient-level predictions are stored only in Git-ignored directories; manuscript tables derive from aggregate artefacts. Reporting will follow TRIPOD+AI and use PROBAST+AI for appraisal [24, 25].

4.2 Clinical question, intended use and target population

The primary question is whether postoperative overall-survival risk can use optional multimodal evidence only when it improves reliability over a clinical-pathological anchor. Prediction occurs immediately after definitive surgery and pathology review. The model is not currently intended to recommend adjuvant therapy. Anchor predictors are age, sex, smoking, site, grade, p16, resection, pathological T and pathological N. Adjuvant treatment, downstream outcomes and post-prediction measurements are excluded.

For patient i , let T_i be event time, C_i censoring time, $Y_i = \min(T_i, C_i)$ observed duration and $\delta_i = I(T_i \leq C_i)$ the event indicator. In HANCOCK,

$$Y_i = \text{days to last information}_i - \text{days to first treatment}_i, \quad (1)$$

$$\delta_i = I(\text{survival status}_i = \text{deceased}). \quad (2)$$

Non-positive durations are excluded. The primary horizon is $t^* = 730.5$ days and the primary prediction is all-cause mortality risk by t^* .

4.3 Cohorts and role separation

HANCOCK is the primary ecosystem for natural missingness and set learning [5]. Its official training subset provides $n = 610$ eligible development patients after one exclusion. The 152-patient official test was outcome-seen earlier and may be evaluated only after freezing, labelled post-lock exploratory.

TCGA-HNSC supplies source RNA-sequencing and clinical outcomes [1, 2]. GSE65858 and GSE41613 supply microarray ecosystems with different case mix and platforms [3, 4]. They study ranking–probability separation, not untouched validation. RADCURE supplies clinical, tumour-volume and CT radiomic analyses [6]. A final cohort must match the postoperative intended use, remain outcome-untouched and have enough events for prespecified estimates. Dataset-specific eligibility, follow-up and ethics language will be reproduced from primary sources.

4.4 Experimental hierarchy and claim gatekeeping

Experiments are organised to answer one primary clinical claim before secondary mechanisms. Stage A freezes eligibility, preprocessing, modality usability and the clinical anchor. Stage B compares C3, HAF, unanchored set models, anchored set models and the complete PATTERN policy by repeated nested out-of-fold prediction at full coverage. Stage C challenges any apparent benefit with natural rare patterns, leave-one-combination-out generalisation, corruption, cross-platform calibration and present-but-invalid controls. Stage D evaluates the fully frozen model and thresholds in an outcome-untouched postoperative cohort.

Claims are gated accordingly. Superiority requires the primary paired Brier difference to favour PATTERN with its interval excluding the frozen no-material-benefit margin. A reliability claim additionally requires worst-pattern clinical regret not to cross the no-harm margin. Selective benefit must persist on identical retained patients and cannot be inferred from reduced coverage alone. Transportability of ranking and transportability of probability are claimed separately. External generalisation is not claimed without Stage D, irrespective of performance in outcome-seen cohorts.

4.5 Modality availability, usability and quality

For optional modality m , we distinguish acquisition a_{im} , usability u_{im} and quality vector q_{im} . Acquisition means a source record or assay exists. Usability means the frozen preprocessing contract can produce valid input. Quality can include outcome-blind fraction missing, out-of-range counts, assay completeness and distance from training distributions. A content token exists only if $u_{im} = 1$; missing modalities do not contribute a zero vector because zero may be a valid observed value.

$$\mathcal{M}_i = \{m : u_{im} = 1\}. \quad (3)$$

The natural pattern is the usability tuple across blood, ICD and TMA. Acquired-but-unusable status is retained for routing. Counts and quality distributions are computed without outcomes.

4.6 Clinical-pathological anchor

The anchor is an elastic-net Cox proportional-hazards model. After fold-specific preprocessing,

$$\eta_{c,i} = f_c(x_{c,i}) = x_{c,i}^\top \hat{\beta}_c. \quad (4)$$

The partial-likelihood objective follows Cox [8]. Twelve prespecified penalty/mixing candidates are selected in three-fold inner cross-validation within five outer folds for seeds 17, 29, 43, 71 and 101. Encoding, imputation, scaling and category handling are training-fold only. A training-fold Breslow estimator gives

$$\hat{S}_c(t | x_{c,i}) = \exp\{-\hat{H}_{0,c}(t) \exp(\eta_{c,i})\}, \quad \hat{R}_c(t) = 1 - \hat{S}_c(t | x_{c,i}), \quad (5)$$

where $\hat{H}_{0,c}$ is cumulative baseline hazard [9]. Table 2 reports repeated out-of-fold estimates.

4.7 Residual set survival backbone

Each usable optional modality is processed by a modality-specific adapter g_m mapping content to 64 dimensions. A token combines content, learned modality identity, acquisition/usability status, quality and allowed domain metadata:

$$z_{im} = g_m(x_{im}) + e_m + h(a_{im}, u_{im}, q_{im}, d_i). \quad (6)$$

The minimum model is Deep Sets [27],

$$h_i = \rho \left(\sum_{m \in \mathcal{M}_i} \phi(z_{im}) \right). \quad (7)$$

The candidate complex model is a lightweight Set Transformer with dimension 64, two attention blocks, four heads, feed-forward dimension 128 and token dropout 0.20 [28]. Escalation is allowed only if the simpler model has a reproducible limitation.

The residual correction is $\Delta\eta_i = r(h_i, \eta_{c,i})$ and

$$\eta_{f,i} = \eta_{c,i} + \Delta\eta_i. \quad (8)$$

The primary objective is negative Cox partial log likelihood. The anchor is initially frozen; joint fine-tuning is sensitivity analysis. If $\mathcal{M}_i = \emptyset$, the implementation must return exactly $\eta_f = \eta_c$ and the same survival curve within numerical tolerance. Unit tests assess fallback, arbitrary subsets, token permutation invariance and exclusion of outcome-derived metadata.

4.8 Survival probability and calibration bridge

A training-fold Breslow baseline maps continuous score to source-domain probability. Target-domain calibration is represented at horizon t as

$$\text{logit } R_{i,t}^* = \alpha_{0,t} + \alpha_{p(i),t} + \alpha_{d(i),t} + \beta_{p(i),d(i),t} \text{logit } R_{i,t}, \quad (9)$$

where $p(i)$ and $d(i)$ denote pattern and domain. Hierarchical shrinkage pulls sparse terms toward a global intercept and slope. Small cells receive intercept-only or global calibration. If minimum support or calibration-uncertainty criteria fail, the router cannot output full probability and chooses RANK-ONLY, FALLBACK or ABSTAIN.

Strategies are compared at 0, 5, 10, 20 and 40 target events: source Breslow, target baseline, intercept-only, intercept and slope, isotonic calibration, and the bridge. Effective sample sizes, events and uncertainty are reported. Primary calibration measures are calibration-in-the-large and slope, supplemented by plots and Brier score.

4.9 Cross-fitted value and reliability router

The router estimates whether fused evidence is expected to improve an individual prediction without using that individual's outcome to create its training label. For

censoring-adjusted proper loss L ,

$$y_{\text{value},i} = L(\hat{R}_{c,i}, Y_i, \delta_i) - L(\hat{R}_{f,i}, Y_i, \delta_i), \quad (10)$$

where both predictions are out of fold. Router covariates include pattern frequency, internal missingness, acquisition/usability mismatch, modality and representation OOD scores, ensemble disagreement, $|\hat{R}_f - \hat{R}_c|$, leave-one-modality sensitivity, calibration uncertainty and shortcut probes. They exclude outcome-derived variables and test-performance summaries.

A frozen policy maps these to four actions:

- FUSE: report calibrated fused probability and ranking when evidence, calibration and value criteria pass;
- FALLBACK: report the anchor probability when optional evidence is absent, unreliable or expected to harm;
- RANK-ONLY: report relative risk or risk group without absolute probability when ranking is supported but calibration is not;
- ABSTAIN: withhold automated output when neither fused nor clinical output meets frozen requirements.

Thresholds are selected only within development/calibration data with complexity penalties and minimum cell support. Official-test, GEO and RADCURE outcomes cannot tune the router.

4.10 Comparators

The benchmark distinguishes gains from representation, residual anchoring, calibration and routing (Table 5). Classical comparators are Kaplan–Meier constant risk, conventional Cox, elastic-net anchor and random survival forest. Fusion comparators are modality-only Cox, concatenation elastic-net Cox, prior residual fusion B6, prior gate B7, cross-fitted score-level late fusion C3 and missing-aware direct fusion C4. Recent incomplete-modality benchmarks include DPLSurv and HAF where input definitions and code permit; HAF is a mandatory HANCOCK comparator because it was developed on the same multimodal resource [13, 14]. Missing-data baselines include zero/mean imputation, missing indicators, modality dropout and separate models for sufficiently large patterns. Deep Sets and Set Transformer are tested with and without residual anchoring. Missingness-only, modality permutation and random-feature controls probe shortcuts.

4.11 Missingness, unseen combinations and corruption

Natural patterns are analysed first. Simulated missingness includes block dropout, missing completely at random at several rates and MAR-like mechanisms conditioned on permitted baseline variables. Unseen-combination experiments exclude one supported pattern while retaining its constituent modalities elsewhere, testing compositional generalisation rather than unseen feature domains.

Table 5 Prespecified models and the question answered by each comparison.

Family	Definition	Scientific question
KM/clinical/V0	Constant risk, Cox, elastic-net clinical-pathological anchor	Does complexity improve on outcome frequency and routine data?
RSF/neural survival	Random survival forest, DeepSurv-like Cox, discrete-time sensitivity head	Are gains generic non-linearity rather than multimodal design?
Direct fusion	Concatenation with imputation, indicators, C4	Does set modelling improve over a fixed vector?
Recent incomplete-modality	DPLSurv and HAF reproduction where feasible	Does PATTERN improve on contemporary task-specific methods?
Late/residual fusion	C3 score-level and prior B6 residual fusion	Does the new backbone beat strong simple/prior comparators?
Pattern-specific	Separate model for sufficiently large patterns	Is compositional sharing better than independent models?
Set models	Deep Sets/Set Transformer with and without anchoring	Are invariance, residualisation and attention necessary?
PATTERN policy	Set backbone, bridge and four-action router	Does conditional use improve full coverage and worst-pattern safety?
Negative controls	Missingness-only, permuted, Gaussian and volume-matched features	Is value specific to intended evidence?

Corruption leaves a modality marked acquired and applies additive Gaussian noise scaled to training dispersion, location/scale shifts, within-patient feature permutation, across-patient modality permutation, Gaussian replacement and, for imaging, volume-matched random controls. Severity levels are fixed without test outcomes. Performance degradation, uncertainty and routing are traced with severity.

4.12 Primary and secondary estimands

The primary full-coverage estimand is

$$\Delta\text{Brier}_{24} = \text{Brier}_{24}(\text{PATTERN policy}) - \text{Brier}_{24}(\text{C3}). \quad (11)$$

Fallback patients contribute clinical risk; negative values favour PATTERN. Co-primary worst-pattern clinical regret is

$$\mathcal{R}_{\text{worst}} = \max_{p \in \mathcal{P}_{\text{supported}}} \{\text{Brier}_{24}(\text{PATTERN}, p) - \text{Brier}_{24}(\text{clinical}, p)\}. \quad (12)$$

Supported patterns are prespecified as at least 30 patients and 10 events unless a more conservative rule is frozen before comparison.

Secondary estimands are Harrell C, Uno C, cumulative/dynamic AUC, integrated Brier, calibration-in-the-large, calibration slope, full-to-subset degradation, seen-unseen gap, error-uncertainty association, actions and risk-coverage. Decision-curve analysis is exploratory potential utility only [26].

4.13 Censoring, metric estimation and inference

IPCW metrics use a censoring distribution estimated from relevant training/reference data with frozen weight truncation. Brier score follows established censoring-adjusted estimation [16]; Uno concordance reduces sensitivity to censoring [17]. Calibration slope uses a horizon-appropriate censoring-aware estimator. Exact software versions will be archived.

Comparisons use 2,000 paired patient-level bootstrap replicates. Patients, not prediction rows, are resampled. For repeated cross-fitting, seed aggregation precedes primary paired intervals so repeated predictions from one patient are not independent. We report effect differences and 95% intervals according to a frozen simulation check. Interpretation is hierarchical: primary full-coverage and worst-pattern estimands, secondary performance/calibration, then ablations/mechanisms. P values, if any, are secondary.

4.14 Selective prediction and matched coverage

Selective performance is reported at 100%, 95%, 90% and 80% coverage and continuously. At each coverage, candidate and comparator are evaluated on the identical retained subset. Coverage, events, pattern composition and clinical subgroup composition are reported together. Insufficient-event subgroups receive uncertainty without definitive fairness claims.

4.15 Ablations and stopping rules

Ablations remove clinical residualisation, quality/status features, pattern terms, domain terms, hierarchical shrinkage, calibration uncertainty, value supervision, leave-one-modality sensitivity, exact fallback and token dropout. Complexity is retained only after reproducible improvement without worse safety regret. Development stops instead of model shopping when C3 matches the set model within a no-material-benefit margin, negative controls match real modality, router value is not associated with error, or calibration cannot be supported at available events.

4.16 Sample-size considerations

Available development size is constrained by public data. Effective information is reported through patients, events, censoring, pattern support and calibration events, with degrees of freedom and shrinkage limited accordingly. Future untouched-validation size will be calculated before outcome access for precision of Brier difference, calibration slope and key pattern estimates, following development and external-validation guidance [22, 23]. Inadequately precise cohorts will be labelled feasibility studies.

4.17 Reproducibility and governance

Fixed seeds are 17, 29, 43, 71 and 101. Cohorts are configuration controlled and each frozen stage has immutable receipts. Two V0 reruns produced identical out-of-fold SHA256 hashes. Patient identifiers and predictions are excluded from version control; only aggregate metrics, manifests and audits are tracked. Each stage records the analysis label, prior outcome access, external-outcome tuning status and frozen-file integrity. New methods use an isolated namespace and cannot modify Phase 3–6 assets.

4.18 Use of generative artificial intelligence assistance

A large-language-model-assisted coding and writing tool was used to help organise the initial manuscript scaffold, edit language and check the LaTeX build. It was not treated as an author and did not independently generate, select or validate study results. All scientific statements, citations, analyses and final wording must be verified and approved by the named authors, who retain full accountability. **[TBD: Before submission, revise this disclosure to match the exact tools, versions, dates and permitted journal wording.]**

4.19 Ethics

This work uses retrospective datasets under original governance and access conditions; no participants were newly recruited. The final submission will reproduce dataset-specific review-board approval, consent, waiver and data-use language from primary publications and repositories. **[TBD: Confirm whether local secondary analysis requires ethics review/exemption and provide identifier.]** No patient/public involvement has been documented for model design; this will be stated if unchanged.

Data availability

HANCOCK, TCGA-HNSC, GSE65858, GSE41613 and RADCURE are available subject to source conditions [1, 3–6]. The final statement will list accessions, versions, download dates and controlled-access boundaries. Patient-level predictions will not enter version control or be redistributed contrary to source terms. Aggregate tables needed for claims will accompany the archived release. **[TBD: Add repository links and accession/version table.]**

Code availability

The configuration-driven `trust-hn` repository contains tests, immutable receipts and aggregate-only reporting. A public release will include preprocessing contracts, model code, frozen configurations, environment locks and scripts regenerating aggregates from lawfully obtained data. **[TBD: Create archival release/DOI; add licence and hardware/runtime.]**

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Author contributions

[TBD: Complete CRediT roles for all authors.]

Competing interests

The authors declare [TBD: no competing interests / specify].

Additional information

Supplementary information: reporting checklist, cohort contract, hyperparameters, model card, calibration plots, pattern estimates, negative controls and reproducibility receipts will be supplied.

Correspondence: [TBD: corresponding author].

Provenance and peer review: [TBD: journal completion].

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